ICT702 Data Visualisation

Purposeful Use of Colour

Module 4

USC AUSTRALIA

I would like to acknowledge the traditional custodians of the land we are meeting on,

The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

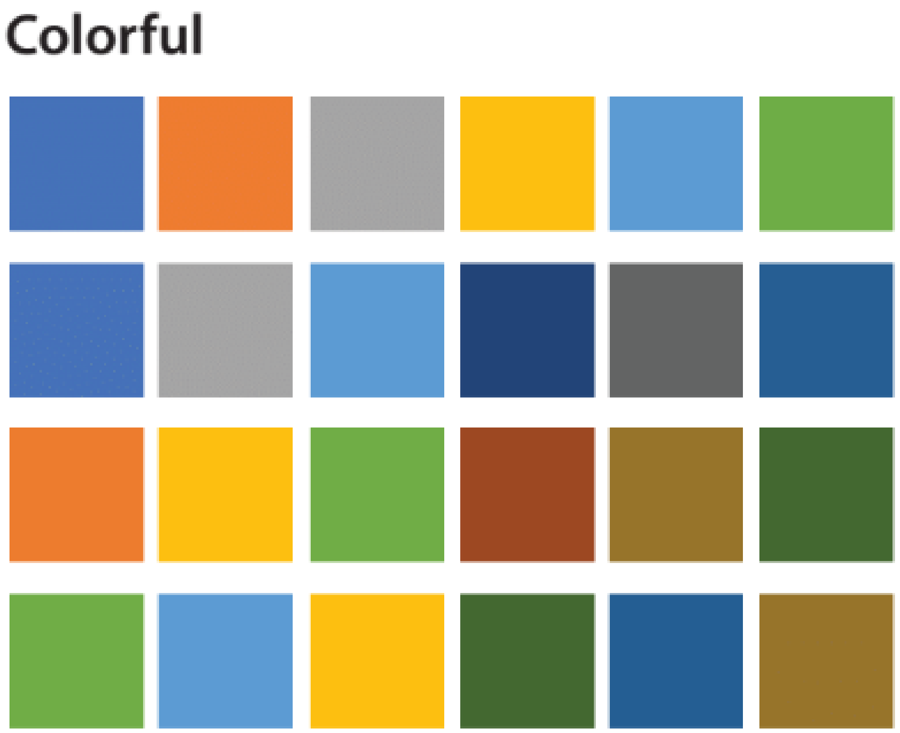
Part 2

Colour Scheme

The **colour scheme** is the set of colours (hues, saturations, and luminances) in data visualization.

The colour scheme depends on the type of data used in the data visualization and the message we want to convey to the audience.

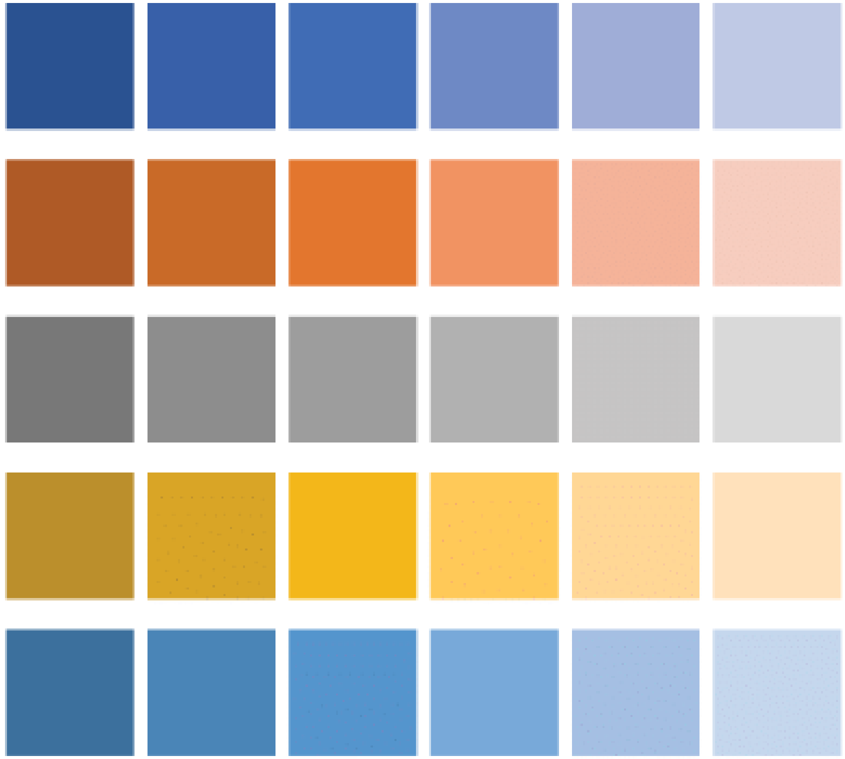
Categorical Colour Schemes



In a **categorical colour scheme**, distinct and unordered colour groups represent a categorical variable's outcomes.

Sequential Colour Schemes

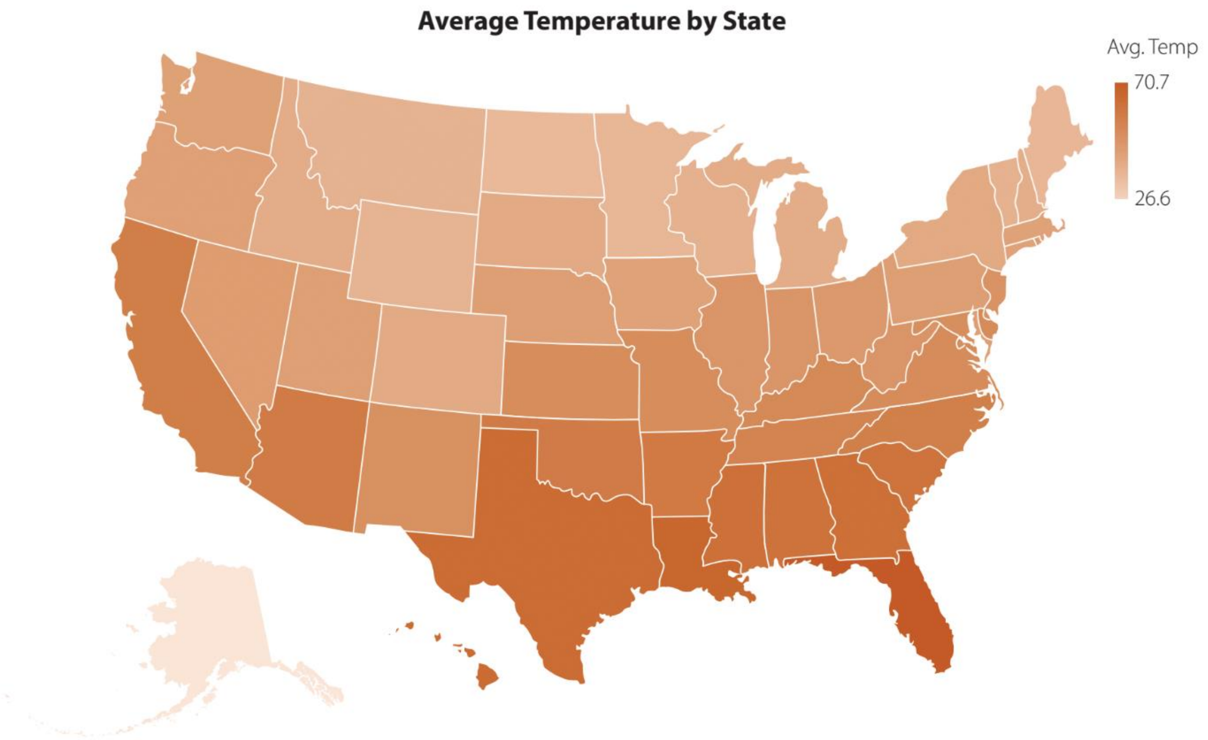
Monochromatic



In a **sequential colour scheme**, the gradient of saturation or luminance of a hue represents the outcomes of an ordered variable.

Sequential Colour Schemes for Maps

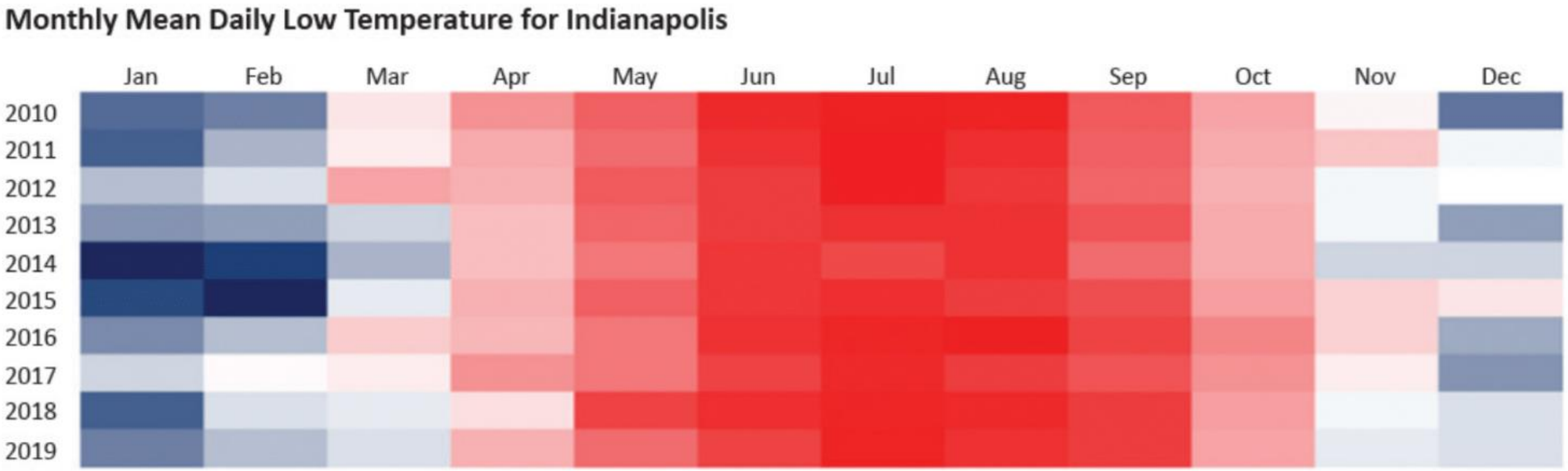
Choropleth map of average annual temperature by state using brown



A monochromatic brown is more suitable to convey an idea of warmth to the audience.

Diverging Colour Schemes

Heat map of monthly mean daily low temperature for Indianapolis 2010–2019



In a **diverging colour scheme**, a gradient formed by two sequential colour schemes sharing a common endpoint represents a quantitative variable.

Each sequential colour scheme, one for values above the reference value and the other one below, uses a different hue with a gradient of increasing luminance as the values approach the reference value.

The HSL Colour System

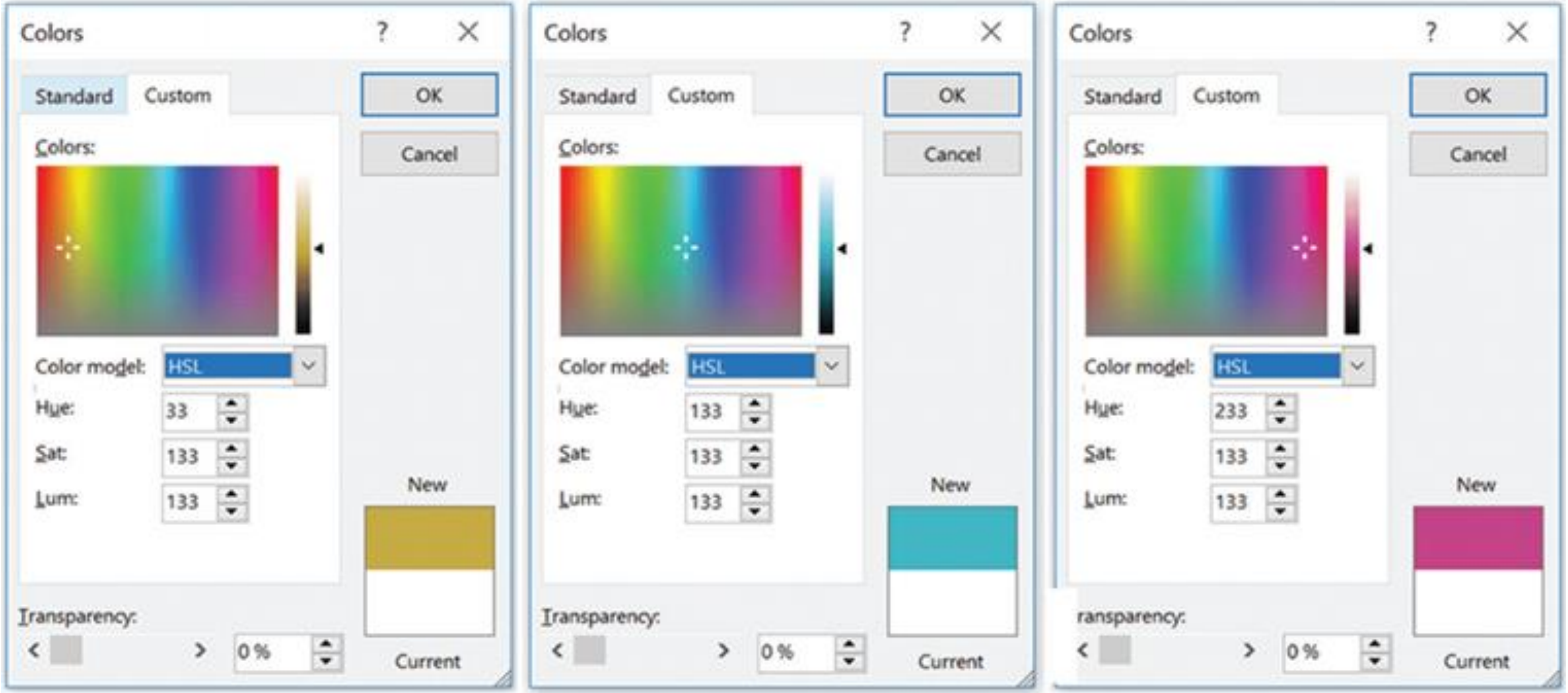
Hue: the primary and secondary colours of the RGB primary colour mode using fixed values for 100% saturation (Sat: 255) and 50% luminance (Lum: 128) are:

Color	Hue
Red	0
Yellow	40
Green	80
Cyan	120
Blue	160
Magenta	200

Sat: a higher saturation represents a purer colour. A Sat value of 0 results in a gray tone.

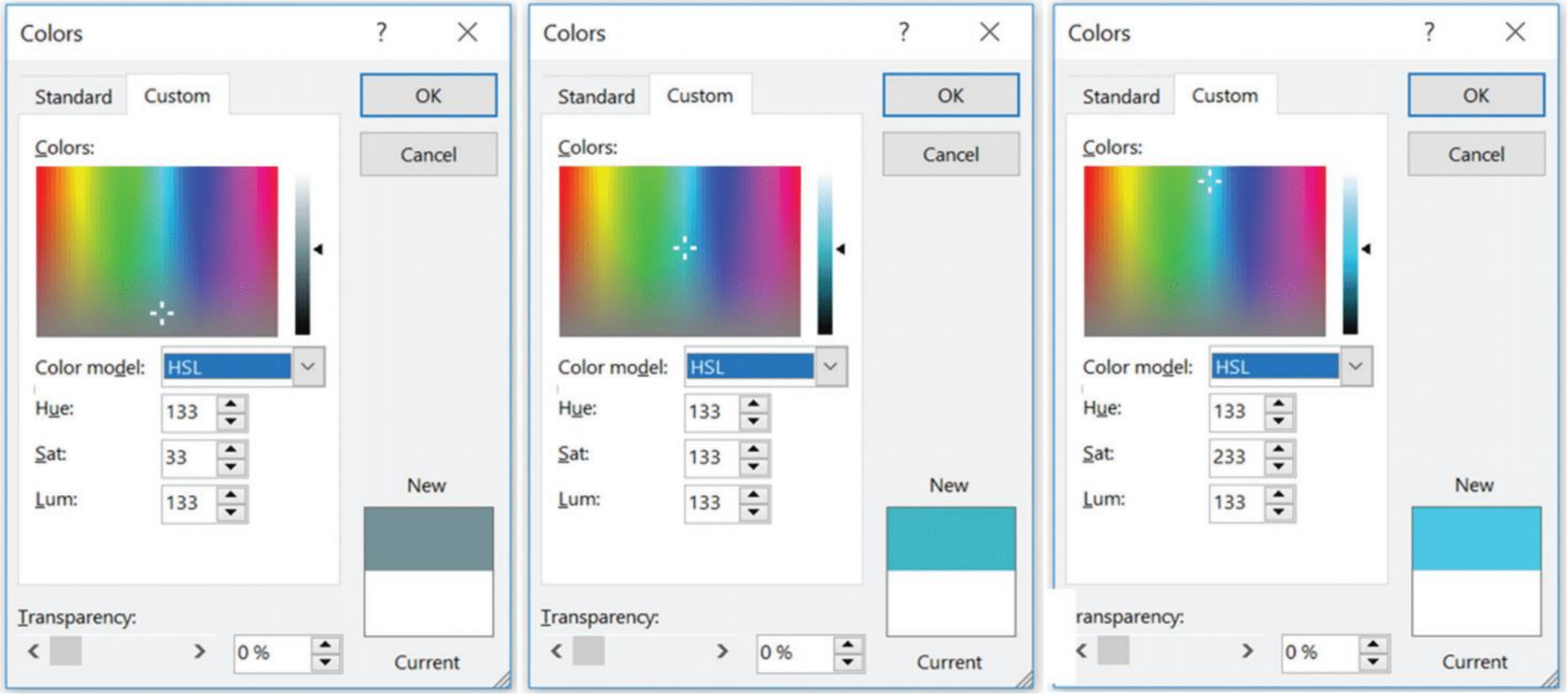
Lum: a luminance of 255 results in the colour white. A Lum value of 0 produces the colour black.

Setting the Hue Parameter in the Colours Dialog Box



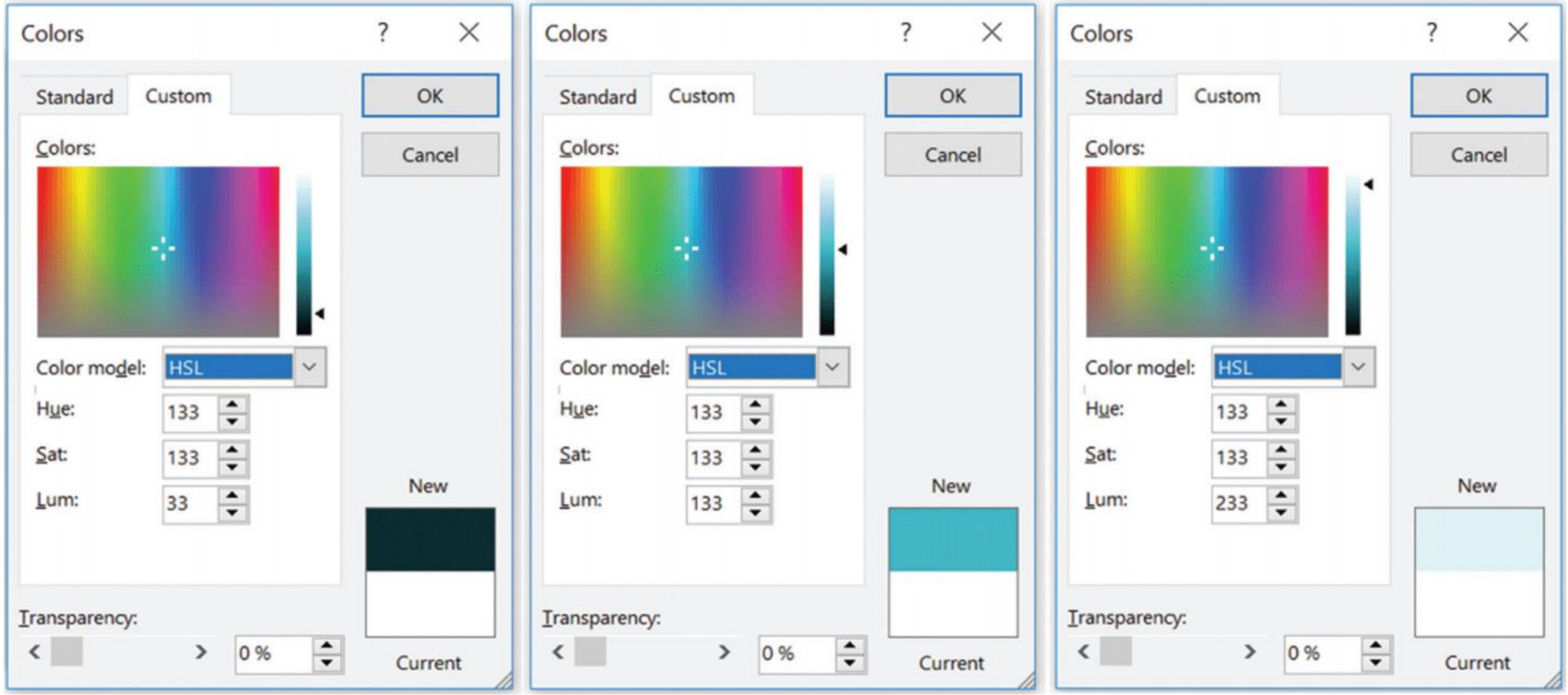
As the value of the **Hue** parameter increases, the crosshair indicator moves horizontally from left to right across the colour spectrum control in the Colours dialog box to indicate the selected hue.

Setting the Sat Parameter in the Colours Dialog Box



As the value of the **Sat** parameter increases, the crosshair indicator moves vertically from bottom to top across the colour spectrum control in the Colours dialog box to indicate the selected saturation, which alters the grayness/increases the purity of the colour.

Setting the Lum Parameter in the Colours Dialog Box

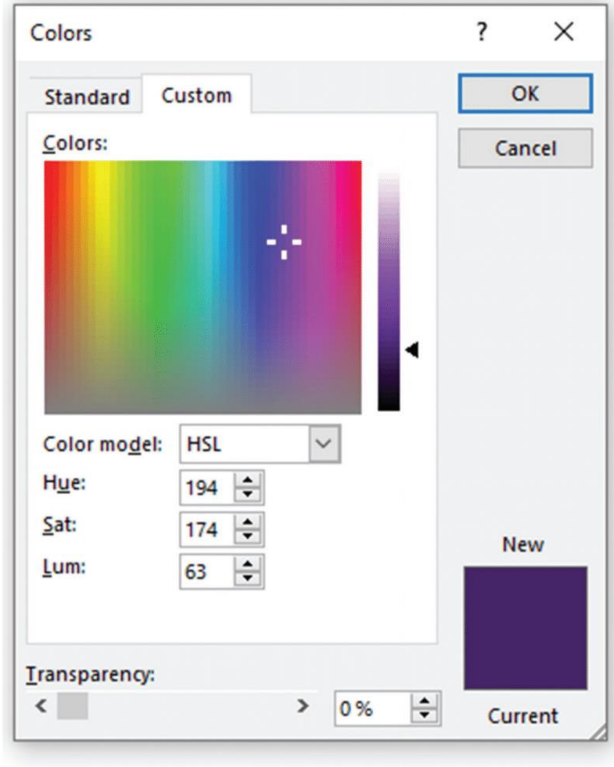


As the value of the **Lum** parameter increases, the ◀ indicator moves vertically from the bottom to the top of the Luminosity slide control in the Colours dialog box to indicate the selected luminance, which reduces and increases the lightness of the colour.

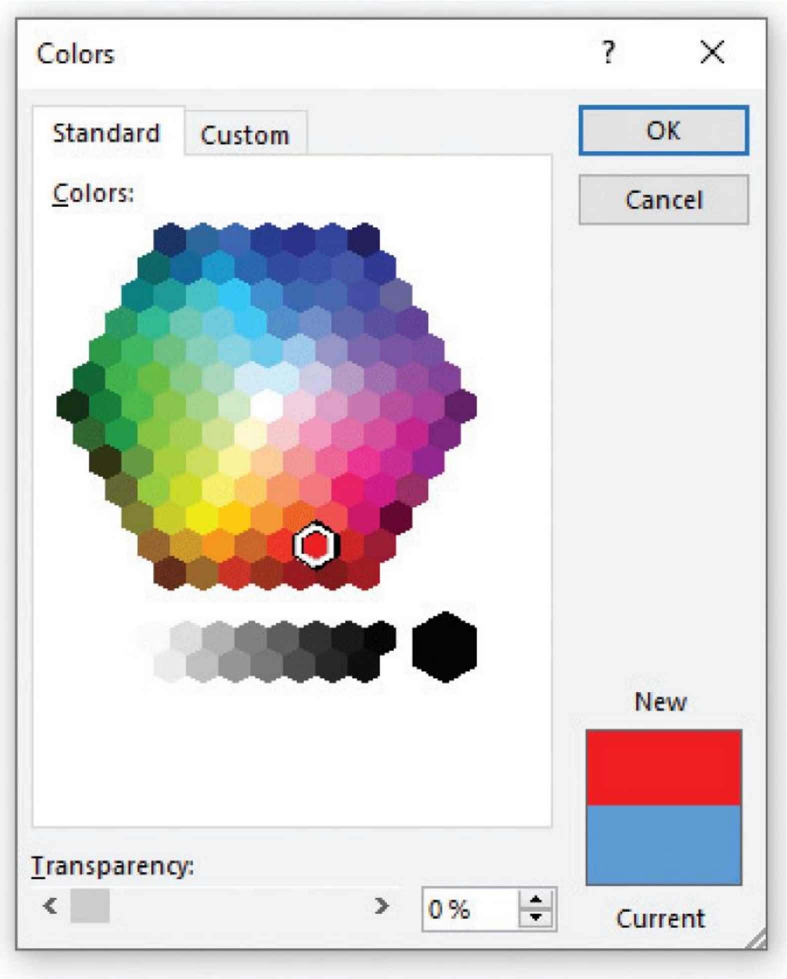
Replicate a Colour Using the Eyedropper Tool

Grappenhall Publishers logo

Hue, Sat, and Lum settings used in the Grappenhall Publishers logo



Custom Colour Using the RGB Colour System



Transparency relates to how much you see-through in a colour.

In Excel the **Transparency** slider can be accessed on the **Standard** tab in the **Colours** dialog box.

Drag the **Transparency** slider or enter a number between 0 and 100 to vary the percentage of transparency.

End of Part 2